

Hands-on Lab: Querying Data in NoSQL Databases

Scenario

You are a data engineer at an e-commerce company. Your company needs you to design a data platform that uses MongoDB as a NoSQL database. You will be using MongoDB to store the e-commerce catalog data.

Assignment Overview

In this assignment, you will perform a series of tasks in a single exercise. But before proceeding with the assignment, you will install the **'mongoimport'** and **'mongoexport'** in the lab environment and then download the **JavaScript Object Notation (JSON)** file. The exercise requires you to first import the JSON file to the MongoDB server into a database and a collection (electronics), then list out all the databases and the collections in the database catalog. Next, you will list the first five documents in the collection, and then you will write a query to find the count of laptops, the number of mobile phones with a screen size of 6 inches, and the average screen size of smartphones. Finally, you will export the ID, type, and model fields from the collection into a Comma Separated Values (CSV) file. After performing each task, you will take a screenshot of the command used, and the output obtained and give a name to the screenshot.

Objectives

In this assignment you will:

- import data into a MongoDB database.
- query data in a MongoDB database.
- export data from MongoDB.

Tools / Software

- MongoDB Server
- MongoDB Command Line Backup Tools

Exercise 1 - Check the lab environment

Before you proceed with the assignment :

- Check if you have the 'mongoimport' and 'mongoexport' installed on the lab, otherwise install them.
- Download the `catalog.json` file from <https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0321EN-SkillsNetwork/nosql/catalog.json>.

- Install Mongo Tools

```
wget https://fastdl.mongodb.org/tools/db/mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz
tar -xf mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz
export PATH=$PATH:/home/project/mongodb-database-tools-ubuntu1804-x86_64-100.3.1/bin
echo "done"
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0321EN-SkillsNetwork/nosql/catalog.json
mongoimport --version
mongoexport --version
echo "installation verified"
start_mongo
mongoimport -u root --authenticationDatabase admin --db catalog --collection electronics --file catalog.json
echo "file catalog.json imported"
mongoexport -u root --authenticationDatabase admin --db=catalog --collection=electronics --type=csv --fields=_id,type,model --out=electronics.csv
echo "file electronics.csv exported"
```

- Check Mongo intall

```

theia@theiadocker-ccmartin:/home/project$ wget https://fastdl.mongodb.org/tools/db/mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz
--2022-03-05 00:48:51-- https://fastdl.mongodb.org/tools/db/mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz
Resolving fastdl.mongodb.org (fastdl.mongodb.org)... 13.225.221.61, 13.225.221.116, 13.225.221.67, ...
Connecting to fastdl.mongodb.org (fastdl.mongodb.org)|13.225.221.61|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 64021650 (61M) [binary/octet-stream]
Saving to: 'mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz'

mongodb-database-tools-ubuntu1804-x86_64-100 100%[=====] 61.06M 29.4MB/s in 2.1s

2022-03-05 00:48:54 (29.4 MB/s) - 'mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz' saved [64021650/64021650]

theia@theiadocker-ccmartin:/home/project$ tar -xvf mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz
theia@theiadocker-ccmartin:/home/project$ export PATH=$PATH:/home/project/mongodb-database-tools-ubuntu1804-x86_64-100.3.1/bin
theia@theiadocker-ccmartin:/home/project$ echo "done"
done
theia@theiadocker-ccmartin:/home/project$ mongoimport --version
mongoimport version: 100.3.1
git version: 32632b931f9c41d8314b75ecc88e551b012b1e30
Go version: go1.15.8
os: linux
arch: amd64
compiler: gc
theia@theiadocker-ccmartin:/home/project$ mongoexport --version
mongoexport version: 100.3.1
git version: 32632b931f9c41d8314b75ecc88e551b012b1e30
Go version: go1.15.8
os: linux
arch: amd64
compiler: gc
theia@theiadocker-ccmartin:/home/project$

```

Exercise 2 - Working with MongoDB

Task 1 - Import 'catalog.json' into mongodb server into a database named 'catalog' and a collection named 'electronics'

```

theia@theiadocker-ccmartin:/home/project$ mongoimport -u root -p NTA00C1jY21hcnRp --authenticationDatabase admin --db catalog --collection electronics --file catalog.json
2022-03-05T14:40:20.449+0000 connected to: mongodb://localhost/
2022-03-05T14:40:20.500+0000 438 document(s) imported successfully. 0 document(s) failed to import.
theia@theiadocker-ccmartin:/home/project$

```

Task 2 - List out all the databases

```

> show databases
admin      0.000GB
catalog    0.000GB
config     0.000GB
local      0.000GB
>

```

Task 3 - List out all the collections in the database `catalog` .

```
> use catalog
switched to db catalog
> show collections
electronics
> █
```

Task 4 - Create an index on the field "type"

```
> db.electronics.createIndex({type: 1})
{
  "createdCollectionAutomatically" : false,
  "numIndexesBefore" : 1,
  "numIndexesAfter" : 2,
  "ok" : 1
}
> █
```

Task 5 - Write a query to find the count of laptops

```
> db.electronics.count({type: 'laptop'})
389
> █
```

Task 6 - Write a query to find the number of **smart phones with screen size of 6 inches.**

```
> db.electronics.count({"screen size": 6})
8
> █
```

Task 7. Write a query to find out the average screen size of **smart phones .**

```
> db.electronics.aggregate([{$group: {_id: "$type", avgSize: {$avg: "$screen size"} }}])
{ "_id" : "television", "avgSize" : 39.8 }
{ "_id" : "laptop", "avgSize" : 14.568123393316196 }
{ "_id" : "smart phone", "avgSize" : 6 }
> █
```

Task 8 - Export the fields **_id, "type", "model", from the 'electronics' collection into a file named**

The screenshot displays an IBM Cloud Lab interface. On the left, a file explorer shows the project structure with files like `catalog.json`, `electronics.csv`, and `install-mongo-tools.sh`. The main terminal window shows the execution of `install-mongo-tools.sh`, which includes downloading MongoDB tools, importing a catalog, and exporting data to `electronics.csv`. The terminal output confirms the successful export of 438 records.

```
1 _id,type,model
2 ObjectId(62238f01ab79830001d57795),smart phone,bn20
3 ObjectId(62238f01ab79830001d57796),smart phone,plk2
4 ObjectId(62238f01ab79830001d57797),smart phone,platina
5 ObjectId(62238f01ab79830001d57798),smart phone,xm23
6 ObjectId(62238f01ab79830001d57799),smart phone,bn20
7 ObjectId(62238f01ab79830001d5779a),smart phone,k9
8 ObjectId(62238f01ab79830001d5779b),smart phone,c3
9 ObjectId(62238f01ab79830001d5779c),smart phone,c12
10 ObjectId(62238f01ab79830001d5779d),smart phone,cz
11 ObjectId(62238f01ab79830001d5779e),smart phone,cz
12 ObjectId(62238f01ab79830001d5779f),laptop,qw2
13 ObjectId(62238f01ab79830001d577a0),laptop,qw2
14 ObjectId(62238f01ab79830001d577a1),smart phone,k9
15 ObjectId(62238f01ab79830001d577a2),laptop,ay9
16 ObjectId(62238f01ab79830001d577a3),laptop,c6
17 ObjectId(62238f01ab79830001d577a4),laptop,k12
18 ObjectId(62238f01ab79830001d577a5),smart phone,xm23
19 ObjectId(62238f01ab79830001d577a6),smart phone,platina
```

```
1 wget https://fastdl.mongodb.org/tools/db/mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz
2 tar -xvf mongodb-database-tools-ubuntu1804-x86_64-100.3.1.tgz
3 export PATH=$PATH:/home/project/mongodb-database-tools-ubuntu1804-x86_64-100.3.1/bin
4 echo "done"
5 wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0321EN-SkillsNetwork/nosql/catalog.json
6 mongoimport --version
7 mongoexport --version
8 echo "installation verified"
9 start_mongo
10 mongoimport -u root --authenticationDatabase admin --db catalog --collection electronics --file catalog.json
11 echo "file catalog.json imported"
12 mongoexport -u root --authenticationDatabase admin --db=catalog --collection=electronics --type=csv --fields=_id,type,model --out=electronics.csv
13 echo "file electronics.csv exported"
```

theia@theiadocker-ccmartin:/home/project\$